

Disproof of Conjecture 00000007788

A lower bound for the missing area of a random polygon

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Abstract

The conjecture predicts an expected volume deficit of order $N^{-2/(n-1)}$ for the convex hull of N independent points uniform in the Euclidean ball. In dimension two, we prove that the deficit is at least $1/[4(N+1)]$. Its quadratic normalization therefore tends to $+\infty$, contradicting the asserted inverse-square asymptotic for every finite leading constant. The accompanying Lean project formalizes the probability law, actual convex-hull area, measurability, integrability, quantitative bound, and asymptotic contradiction.

1 Statement and scope

Both language versions define $K_N = \text{conv}(X_1, \dots, X_N)$, where the X_i are independent and uniform on K . For a solid Euclidean ball, this is normalized volume on the ball. The later reference to spherical measure does not change that sampling law. Dimension $n = 2$ is an admissible instance of the displayed exponent.

Let $B = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$, let λ be Lebesgue area, and set

$$\mu(A) = \frac{\lambda(A \cap B)}{\pi}, \quad H_N(x) = \text{conv}\{x_i : 1 \leq i \leq N\}, \quad D_N = 1 - \frac{\mathbb{E}_{\mu^{\otimes N}} \lambda(H_N)}{\pi}.$$

The necessary ball assertion of the conjecture gives $D_N \sim c_2 N^{-2}$ for a finite positive constant c_2 . Refuting this necessary assertion disproves the original conjunction; it requires no choice of definitions for its additional constant-ratio or simplex corner-measure clauses.

Theorem 1. *For every integer $N \geq 1$,*

$$D_N \geq \frac{1}{4(N+1)}.$$

Consequently $N^2 D_N \rightarrow +\infty$. In particular, $N^2 D_N$ has no finite real limit, and $D_N \not\sim c N^{-2}$ for every $c > 0$.

2 Measurability and existence of the expectation

For completeness, the expectation above is an ordinary expectation of an integrable random variable. For each fixed N , put

$$\Delta_N = \{w \in \mathbb{R}^N : w_i \geq 0, \sum_i w_i = 1\}.$$

This simplex is compact, and $H_N(x) = \{\sum_i w_i x_i : w \in \Delta_N\}$. The set of (x, y, w) with $w \in \Delta_N$ and $y = \sum_i w_i x_i$ is closed by continuity of the finite sum. Projection along the compact factor Δ_N is a closed map, so

$$G_N = \{(x, y) : y \in H_N(x)\}$$

is closed. The measurable-section theorem for product measurable spaces shows that $x \mapsto \lambda(H_N(x))$ is measurable. Each finite hull is compact and has finite area. Under $\mu^{\otimes N}$ all sample points lie in B almost surely, and convexity gives $H_N \subseteq B$. Thus $0 \leq \lambda(H_N)/\pi \leq 1$ almost surely. Both normalized and unnormalized hull area are integrable. The formalization proves these facts explicitly before using integral monotonicity.

3 The elementary lower bound

For $0 \leq r \leq 1$, let $E_r = \{x : \|x_i\| \leq r \text{ for all } i\}$. Area scaling and independence give

$$\mu(rB) = r^2, \quad \mathbb{P}(E_r) = (r^2)^N.$$

On E_r , convexity gives $H_N \subseteq rB$, hence $\lambda(H_N)/\pi \leq r^2$. Off this event the almost-sure bound by 1 remains valid. Therefore

$$\frac{\lambda(H_N)}{\pi} \leq 1 - (1 - r^2)1_{E_r} \quad \text{almost surely.}$$

Taking expectations and rearranging proves

$$D_N \geq (1 - r^2)(r^2)^N. \tag{1}$$

Choose

$$t = \frac{1}{2(N+1)}, \quad r = \sqrt{1-t}.$$

Here $0 < t \leq 1/2$ and $0 \leq r \leq 1$. Bernoulli's inequality yields

$$(r^2)^N = (1-t)^N \geq 1 - Nt \geq \frac{1}{2}.$$

Substituting into (1) gives $D_N \geq t/2 = 1/[4(N+1)]$. For $N \geq 1$ it follows that

$$N^2 D_N \geq \frac{N^2}{4(N+1)} \geq \frac{N}{8} \rightarrow +\infty.$$

For each $c > 0$, the ratio $D_N/(cN^{-2}) = N^2 D_N/c$ also tends to $+\infty$, and cannot tend to 1. This proves Theorem 1. The bound is sufficient for a disproof and makes no claim about the sharp asymptotic order of the deficit.

4 Correspondence with the Lean project

The project uses Lean 4.19.0 and Mathlib v4.19.0, pinned by the supplied toolchain and dependency manifest. The mathematical modules are `Conjecture7788.lean`, `Conjecture7788/HullMeasurable.lean`, and `Conjecture7788/GenericRate.lean`.

The type `Plane` is `EuclideanSpace` over $\mathbb{R}$ with two coordinates. The definition `disk` is the closed unit ball. `uniformDisk` is Mathlib’s normalized restriction `ProbabilityTheory.cond` applied to `volume disk`; its probability-measure instance follows from the disk area $\pi > 0$. `samples N` is the finite product of these probability measures. The lemma `samples_rectangle` records the exact product law.

`hullArea` is the actual Lebesgue measure of the convex hull of the sample range, converted to a real number and divided by π . The project proves finiteness of every such hull measure, its measurability, and integrability of both area functions. In particular, the use of real-valued integration does not rely on a default value for a nonintegrable function. The lemma `deficit_eq_expected_volume_deficit` identifies `deficit N` with the source’s exact normalized expected-volume deficit.

`deficit_lower_from_radius` proves (1); `deficit_lower` specializes the radius and proves the bound. The generic sequence lemmas have their lower-bound hypothesis discharged for this exact geometric deficit. The final declarations establish quadratic divergence, absence of every finite normalized limit, divergence of the ratio to every positive candidate, and `conjecture_00000007788_false`, which negates existence of a positive constant with the asserted asymptotic equivalence. The real-power exponent $-2/(2 - 1) = -2$ is checked explicitly.

5 Reproduction and verification scope

The submission includes the complete project, `Audit.lean`, a declaration inventory, source hashes, and the portable `verify.py` script. With the pinned dependencies and Lean tools available, run `python3 verify.py` from the `lean` directory. It checks the source hashes and all nine dependency revisions, rebuilds the authored project, replays every authored Lean file with warnings treated as errors, and inspects the types and axiom dependencies of all named declarations. Raw command results and the verification summary accompany the submission.

There are no auxiliary numerical experiments or finite searches in this proof: the quantitative and limiting arguments are symbolic. Only Lean’s standard logical axioms `propext`, `Classical.choice`, and `Quot.sound` are permitted by the audit. Local verification and independent review are documented separately from any later maintainer decision.